UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 7 printed pages.

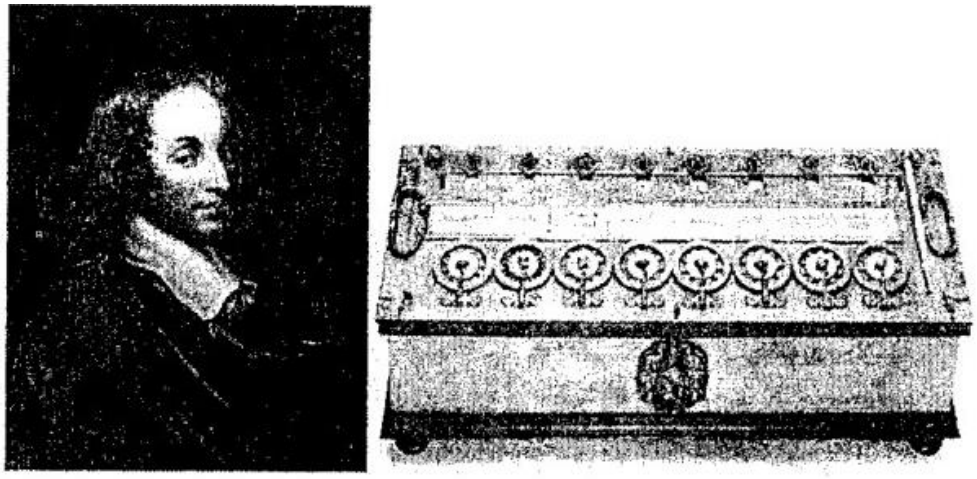
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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

Question 1 is based on the following Figure 1.



Inventor 'Y' was France's most celebrated mathematician and physicist. In 1642, inventor 'Y' invented and build the first working calculating machine as a means of helping his father to perform tedious tax accounting.

a) Who is inventor 'Y'?

b) Name another calculating machine that was invented in the same generation as inventor Y's calculating machine.

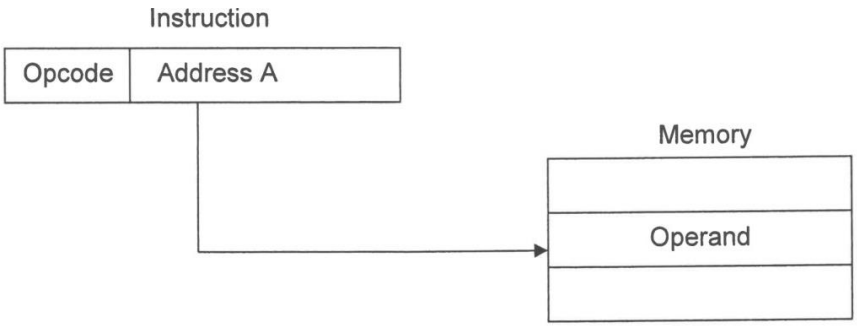
c) Which programming language is named after inventor 'Y'?

d) State TWO (2) mathematical operationsthat can be performed byinventor Y's calculating machine.

QUESTION A2 **(5 marks)**

Given the following Figure 2.

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a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

QUESTION A3 **(5 marks)**

Draw the complete machine cycle and explicitly describe the function of each phase and the location of its occurrence.

QUESTION A4 **(5 marks)**

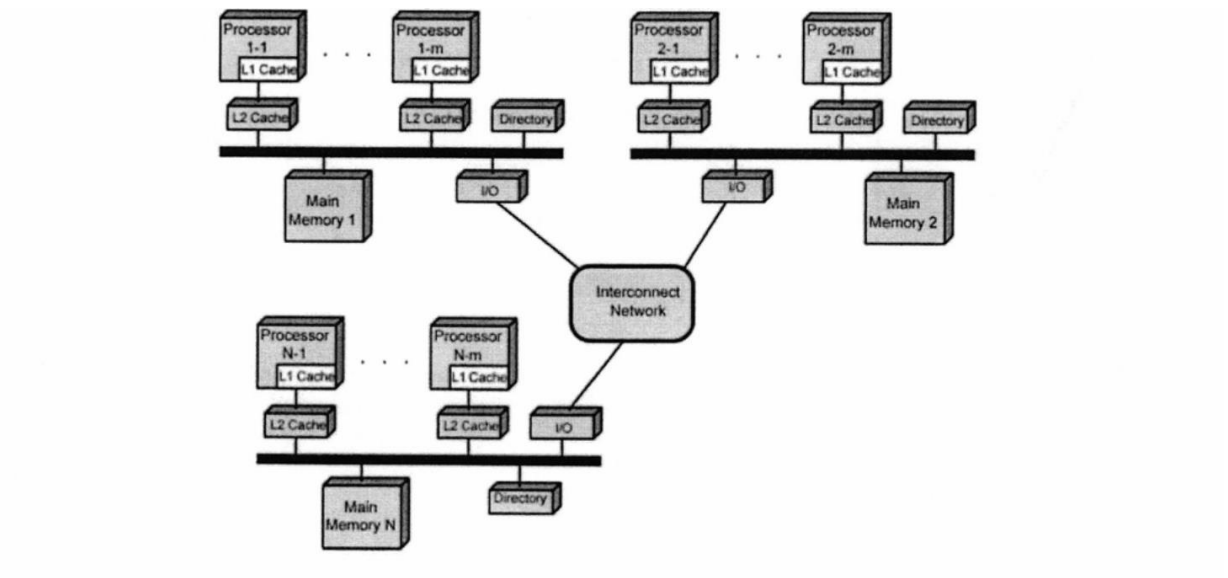
a) What is mapping function?

b) Differentiate between direct and associative mapping in cache.

c) If a 28-bit address is used with word size = 4 bit and line = 10 bit, calculate the tag size of both direct and associative mapping.

QUESTION A5 **(10 marks)**

Question 10 is based on Figure 2. Figure 2 depicts a Non-uniform Memory Access (NUMA) architecture.



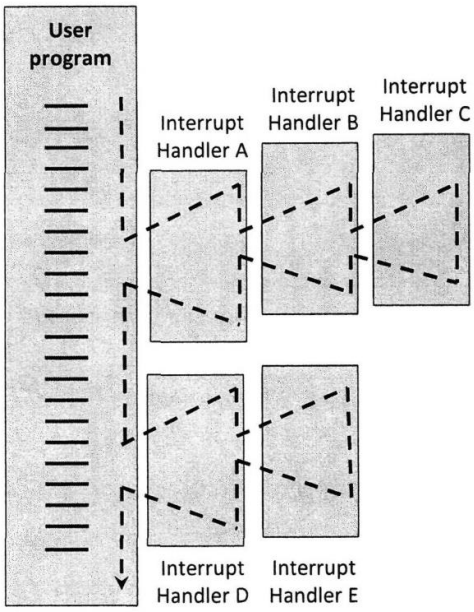
a) Redraw a symmetric multiprocessor organization based on Figure 2.

b) Give TWO (2) examples of tightly coupled Multiple Instruction, Multiple Data (MIMD) parallel processor architectures.

c) Discuss TWO (2) disadvantages of tightly coupled Multiple Instruction, Multiple Data (MIMD) parallel processor architectures.

QUESTION A6 **(10 marks)**

Question 8 is based on Figure 2.



a) Describe maskable interrupt.

b) What is the sequence of the program execution if the defined priority is implemented in

c) What is the sequence of the program execution if the disable interrupt is implemented in

d) Which of the interruption method between defined priority or disable interrupt can finish Interrupt C earlier? Give your justification.

QUESTION A7 **(10 marks)**

a) Why replacement algorithm for cache is needed?

b) Discuss all different types of available replacement.

QUESTION A8 **(10 marks)**

a) When is an I/O module installed?

b) What will happen if the I/O module is not installed?

c) Explain ONE (1) category of I/O module

d) What does it mean by 'hot connection'?

e) List THREE (3) common interfaces to external devices.

QUESTION A9 **(10 marks)**

a) Convert -176.812510 to floating point number in IEEE single precision format in hexadecimal value form.

b) Convert 3FOOC00016 in IEEE single precision floating point number to decimal value form.

QUESTION A10 **(10 marks)**

Keyboard is the most common input/output device to communicate between computer and user.

a) What is the input/output module category for keyboard?

b) State TWO (2) other devices that are in the similar input/output module category?

c) What is the basic unit of exchange for keyboard?

d) Discuss TWO (2) types of characters.

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 1 and Table 1.

$$z = A^2 - B * C + \frac{2}{D + 5}$$

write an appropriate program using correct assembly language notations to compute the equation snbwn in Figure 1 using instructions for zero-address, one-address, and three- address. The available instructioni are provided in Tabte 1. The register labeled A, B, C, and D are fixed, white registers x, y, and z can be used temporarily. The final answer must be stored in register z.

Zero-address	One-address	Three-address
PUSH p	LOAD p	ADD (p,q,r)
POP p	STOR p	suB (p,q,r)
ADD	ADD p	MUL (p,q,r)
SUB	SUB p	DIV (p,q,r)
MUL	MUL p	
DIV	DIV p	